



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M. Tech (CCT) (CBCS) Degree Examinations
May-June 2025

22CCT201 – CONSTRUCTION QUALITY AND SAFETY MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

Unit – I		Marks	BT*	CO*	PO*
1. a)	Analyze the potential challenges and benefits of using quality circles in a construction environment.	10	L*4	1	1,3,4
b)	Identify potential challenges in implementing a quality management system and suggest solutions.	10	L4	1	1,3,4
2. a)	Explain the importance of clearly defined responsibilities and authority in ensuring construction quality.	10	L3	1	1,3,4
b)	Analyze the stages of a quality audit process.	10	L4	1	1,3,4
Unit - II					
3. a)	Discuss the relationship between different construction methods and their impact on achieving various quality objectives.	10	L3	2	1,2,3
b)	Analyze the relationship between project scheduling techniques and the ability to maintain quality while meeting time of completion targets.	10	L4	2	1,2,3
4. a)	Explain the costs and benefits of implementing ergonomic improvements on a construction site.	10	L2	2	1,2,3
b)	Explain the impact of different contract types on the approach to quality management in construction.	10	L3	2	1,2,3
Unit – III					
5. a)	Analyze the potential conflicts between owner objectives and contractor objectives regarding quality.	10	L4	3	1,2,3
b)	describe the factors that influence the reliability of a building component.	10	L2	3	1,2,3
6. a)	Analyze how the quality of materials affects the stability of a structure.	10	L4	3	1,2,3
b)	Analyze the trade-offs between initial cost and long-term maintenance costs in a building project.	10	L4	3	1,2,3
Unit – IV					
7. a)	Explain the root causes of common construction accidents.	10	L2	4	1,2,3
b)	Investigate the factors that led to a case of lead poisoning among construction workers.	10	L4	4	1,2,3
8. a)	Analyze the factors contributing to excavation accidents.	10	L4	4	1,2,3
b)	Determine the root cause of an equipment failure.	10	L4	4	1,2,3
Unit – V					
9. a)	Differentiate between random and systematic safety sampling techniques.	10	L4	5	2,4,6
b)	Determine the adequacy of first-aid facilities on a construction site.	10	L4	5	2,4,6
10. a)	Analyze the relationship between hazard identification and cost reduction.	10	L4	5	2,4,6
b)	Describe challenges of achieving a zero-accident goal in a high-risk industry.	10	L3	5	2,4,6

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
